Taking Internet connectivity—and progress—to remote Indian villages (Source: AirJaldi)

bet X to use FSOC for providing connectivity to remote areas in the state. This initiative syncs with the Andhra Pradesh government's AP Fiber Grid programme that aims to connect 12 million households as well as thousands of businesses to the Internet by next year.

X Labs will provide 2000 FSOC links and required technical support to deploy them. Each FSOC link will provide 20Gbps wireless Internet connection up to a distance of 20km. FSOC links will help bridge the gaps between major access points like cell-towers and Wi-Fi hotspots, thereby servicing thousands of people each.

... And drones

Facebook hopes to achieve a similar feat by using solar-powered drones. Colloquially referred to as a drone, Aquila is more of an advanced aircraft that weighs over 450kg and has a longer wingspan than a Boeing 747. Its wings are made of cured carbon fibre, which enables accurate moulding to achieve the unique shape of wings.

The aircraft will carry communications equipment as the payload. Signals beamed from it will be received by small towers and dishes that will convert them into Wi-Fi or LTE communication signals. A couple of aircrafts positioned properly will work together to provide connectivity to remote areas, extending the reach of existing telecom networks without setting up new towers, laying cables, etc.

Much to the delight of Facebook's UK-based aerospace lab, Aquila successfully completed a test flight in Arizona in mid-2017. Following a failed attempt that crash landed in December 2016, the team improved the drone by fitting more sensors, new spoilers, a horizontal propeller stopping system and so on. These improvements helped the drone to stay in the air for close to two hours, reaching a height of roughly 915 metres (3000 feet).

The challenge before Facebook is to make the aircraft stay in the air for months, something that nobody has done so far! They hope that ultimately their solar-powered aircraft will stay up at a height of 18.3-27km (60,000-90,000 feet) for at least three months, providing communication to a 50km communications radius using free space laser communications technology.

In a press report following the successful test, Mark Zuckerberg of Facebook said: "We successfully gathered a lot of data to help us optimise Aquila's efficiency. No one has ever built an unmanned airplane that will fly for months at a time, so we need to tune every detail to get this right."